

Anik Halder

(he/him/his)

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Education

- Feb 2024 **PhD Physics**, (*magna cum laude*) *Ludwig-Maximilians-Universität München*, Germany.
March 2020 **MSc Astrophysics**, (*distinction*) *Ludwig-Maximilians-Universität München*, Germany.
June 2017 **BSc Physics**, (*distinction*) *Jacobs University Bremen*, Germany.
May 2014 **High school graduation (AISSCE)**, *Hem Sheela Model School*, Durgapur, India.

Employment

University of Cambridge, United Kingdom

- Nov 2024 - present **Research Associate**, *Institute of Astronomy & Kavli Institute for Cosmology, Cambridge*.
- Conducting research projects in gravitational lensing and galaxy formation and evolution (ERC-funded; PI: Prof. Hiranya Peiris).
 - Lead author of the first principled Bayesian inference of galaxy properties for a wide-area weak lensing survey: 4 million KiDS-1000 galaxies with the pop-cosmos generative model.
 - IoA Colloquium and KICC CMB-LSS seminar organiser.
 - Postdoctoral Associate at Jesus College, Cambridge contributing to academic and college events.

Ludwig-Maximilians-Universität München, Germany

- Jan 2019 - **Teaching and Research Assistant**, *University Observatory Munich & Max Planck Institute for Extraterrestrial Physics, Garching*.
- Oct 2024
- Conducting research projects in weak gravitational lensing cosmology.
 - Supervision of student projects, labs and courses.
 - Extragalactic Astronomy research group seminar organiser.

Heinz Maier-Leibnitz Research Centre, Garching, Germany

- Nov 2017 - **Software Developer**, *Scientific Computing group*.
- Dec 2018
- Developer of the small-angle x-ray scattering software BornAgain.

Internships

Instituto de Astrofísica de Canarias, Tenerife, Spain

- June 2017 - **DAAD RISE Weltweit Scholar 2017**.
- Aug 2017
- Quantifying the demographics of Boxy/Peanut structures in edge-on galaxies in the local Universe.

University of St Andrews, United Kingdom

- June 2016 - **DAAD RISE Weltweit Scholar 2016**.
- Aug 2016
- Analysis and validation of realistic synthetic observations of star-forming clouds.

Fraunhofer Institute for Laser Technology, Aachen, Germany

- July 2015 - **Department of Lasers and optics**.
- Aug 2015
- Model validation of thermal surface deformation of optical lenses due to laser beams.

Awards

- Dec 2022 Awarded the Early Career Scientist Travel Grant from the Dark Energy Survey (DES) to attend collaboration-wide meeting. Also invited to present research in the Special Session on New Results from DES at the 241st American Astronomical Society Meeting, Seattle, USA.
- Mar 2022 Awarded a grant from the DAAD RISE Germany programme to host an international undergraduate student for a summer research internship.
- Jun 2017 Placed on the President's List of Jacobs University Bremen in recognition of academic performance during undergraduate study.
- 2016–2017 Awarded the DAAD RISE Weltweit Scholarships in two consecutive years for conducting research projects in the natural sciences at institutions outside Germany.
- 2014–2017 Awarded a merit-based scholarship for undergraduate studies at Jacobs University Bremen, Germany.
- Jun 2014 Awarded the INSPIRE Scholarship for Higher Education (ranked in the top 1% nationally in the All India Senior Secondary Examination, India); *declined in favour of pursuing undergraduate studies in Germany.*
- Sep 2013 Runner-up in the 21st Prof. Brahm Prakash Memorial Materials Essay and Elocution Competition, Indian Institute of Metals Kalpakkam, among 6 finalists from all over India.

Selected talks

Physical properties and redshift distributions of KiDS-1000 galaxies using pop-cosmos

- Oct 2026 **University of Edinburgh, UK** (invited, upcoming).
- Sep 2026 **LSST@Europe8, Budapest, Hungary** (invited, upcoming).
- Sep 2026 **Princeton University, USA** (invited, upcoming).
- Sep 2026 **Stony Brook University, USA** (invited, upcoming).
- Sep 2026 **Astrophysics and Space Science in Marche III: Big Bang, Big Stars, Big Computers, Jesi, Italy** (invited).
- Aug 2026 **COSMO-26, Leiden University, Netherlands.**
- June 2026 **University of Oxford, UK** (invited).
- May 2026 **University of Nottingham, UK** (invited).
- Mar 2026 **UK Cosmology Meeting, Imperial College London, UK.**
- Mar 2026 **Institute of Astronomy Seminar, University of Cambridge, UK.**

Earlier talks

- May 2025 *Higher-order weak lensing analyses and the road to galaxy-population split statistics*, **MIAPbP program “Big Data, Big Questions: The Future of Cosmological Surveys”, Garching, Germany.**
- Dec 2024 *Random matrix theory insights into the precision matrix expansion*, **Astrophysics Group Meeting, Imperial College London, UK.**
- May 2024 *The Integrated 3-point correlation function of projected cosmic density fields*, **18th Kosmologietag, Bielefeld, Germany.**
- April 2023 *The Integrated 3-point correlation function of projected cosmic density fields*, **Future Cosmology summer school, Cargese, France.**
- Feb 2023 *The Integrated 3-point correlation function of cosmic shear*, **Astromerique Speaker Series, University of Montreal, Canada (remotely).**
- Jan 2023 *Response approach to the Integrated shear 3-point correlation function: impact of baryonic effects on small scales*, **Cosmo-Exgal seminar, University College London, UK.**

Jan 2023 *Response approach to the Integrated shear 3-point correlation function: impact of baryonic effects on small scales*, **Special Session on New Results from the Dark Energy Survey, 241st American Astronomical Society Meeting, Seattle, USA.**

2022 *Response approach to the Integrated shear 3-point correlation function: impact of baryonic effects on small scales*, **Institute for Advanced Study, Princeton, USA; Yukawa Institute for Theoretical Physics, Kyoto University, Japan; German Centre for Cosmological Lensing, Ruhr University Bochum, Germany (remotely).**

Other invitations

Sep 2026 Invited external participant, **Simons Collaboration on Learning the Universe Annual Meeting, New York, USA.**

Teaching experience

Oct 2019 - Tutor, grader, and acting lecturer at LMU Munich for BSc Physics and MSc Astrophysics
Oct 2024 courses. Designed and introduced the *Weak Gravitational Lensing* MSc Astrophysics lab.
Sep 2015 - Tutor and grader at Jacobs University Bremen for BSc Physics courses: *Classical Physics*,
June 2017 *Modern Physics, Statistical Physics, Renewable Energy.*

Mentoring experience

Current **Dorsa Sadat Hosseini** (PhD student at the Perimeter Institute for Theoretical Physics, Canada), research project on the clustering of quenched and star-forming DESI luminous red galaxies.
Summer 2026 **Andrew Cunningham** (undergraduate student at the University of Edinburgh), IoA summer student: hierarchical inference of the stellar mass function in KiDS.
April 2023 - **David Gebauer** (currently graduate student at the University of Bielefeld) co-supervision of
present master's and PhD projects.
June 2022 - **Yue Pan** (currently graduate student at Princeton University), DAAD RISE Germany summer
Aug 2022 intern at LMU Munich.
April 2020 - **Zhengyangguang Gong** (currently postdoctoral researcher at the University of Arizona),
July 2025 co-supervision of master's and PhD projects.

Service

Organiser Splinter session *Beyond 2-Point Large Scale Structure Cosmology*, Annual Meeting of the Astronomische Gesellschaft, Garching (Sep 2026)
Reviewer JCAP (Journal), DES (Collaboration), LSST DESC (Collaboration)
Social work Royal Voluntary Service (Cafe retailer)

Skills

Languages Bengali (native), English (bilingual), German (intermediate), Hindi (intermediate).
Computing c, c++, python, high-performance computing

Publications

Selected publications with major contributions. A full publication list, including other collaborative works, is available on my Google Scholar page.

A. **Halder**, H. V. Peiris, S. Thorp, B. Leistedt, D. J. Mortlock, G. Jagwani, M. N. Tudorache, S. Deger, B. Van den Bussche, J. Leja, and A. H. Wright, “pop-cosmos: Redshifts and physical properties of KiDS-1000 galaxies,” *MNRAS* **551** no. 4, (Sept., 2026) [stag1590](#), [arXiv:2602.03930](#) [astro-ph.GA].

B. Leistedt, H. V. Peiris, A. **Halder**, S. Thorp, D. J. Mortlock, A. Loureiro, J. Alsing, G. Jagwani, M. N. Tudorache, S. Deger, J. Leja, B. Van den Bussche, A. H. Wright, S.-S. Li, K. Kuijken, and H. Hildebrandt, “pop-cosmos: Forward modeling KiDS-1000 redshift distributions using realistic galaxy populations,” *arXiv e-prints* (Feb., 2026) [arXiv:2602.03935](#), [arXiv:2602.03935](#) [astro-ph.CO].

D. Gebauer, A. **Halder**, S. Seitz, and D. Anbajagane, “SBi3PCF: Simulation-based inference with the integrated 3PCF,” *J. Cosmology Astropart. Phys.* **2026** no. 6, (June, 2026) 036, [arXiv:2510.13805](#) [astro-ph.CO].

L. Lucie-Smith, H. V. Peiris, A. Pontzen, A. **Halder**, J. Schaye, M. Schaller, J. Helly, R. J. McGibbon, and W. Elbers, “Cosmological feedback from a halo assembly perspective,” *Phys. Rev. D* **112** no. 6, (Sept., 2025) 063541, [arXiv:2505.18258](#) [astro-ph.CO].

J. Harnois-Déraps and the LSST Dark Energy Science Collaboration (incl. A. **Halder**), “Non-linear infusion of intrinsic alignment and source clustering: impact on non-Gaussian cosmic shear statistics,” *MNRAS* **550** no. 2, (July, 2026) [stag1213](#), [arXiv:2509.25166](#) [astro-ph.CO].

O. Friedrich, L. Castiblanco, A. **Halder**, and C. Uhlemann, “Bye binormal: analysing the joint PDF of galaxy density and weak lensing convergence,” *MNRAS* **545** no. 4, (Jan., 2026) [stag2181](#), [arXiv:2507.16957](#) [astro-ph.CO].

Z. Gong, A. **Halder**, A. Bohrdt, S. Seitz, and D. Gebauer, “C3NN: Cosmological Correlator Convolutional Neural Network an Interpretable Machine-learning Framework for Cosmological Analyses,” *ApJ* **971** no. 2, (Aug., 2024) 156, [arXiv:2402.09526](#) [astro-ph.CO].

A. Barthelemy, A. **Halder**, Z. Gong, and C. Uhlemann, “Making the leap. Part I. Modelling the reconstructed lensing convergence PDF from cosmic shear with survey masks and systematics,” *J. Cosmology Astropart. Phys.* **03** (Mar., 2024) 060, [arXiv:2307.09468](#).

A. **Halder**, Z. Gong, A. Barreira, O. Friedrich, S. Seitz, and D. Gruen, “Beyond 3×2 -point cosmology: the integrated shear and galaxy 3-point correlation functions,” *J. Cosmology Astropart. Phys.* **2023** no. 10, (Oct, 2023) 028, [arXiv:2305.17132](#).

Z. Gong, A. **Halder**, A. Barreira, S. Seitz, and O. Friedrich, “Cosmology from the integrated shear 3-point correlation function: simulated likelihood analyses with machine-learning emulators,” *J. Cosmology Astropart. Phys.* **2023** no. 7, (July, 2023) 040, [arXiv:2304.01187](#).

A. **Halder** and A. Barreira, “Response approach to the integrated shear 3-point correlation function: the impact of baryonic effects on small scales,” *MNRAS* **515** no. 3, (Sept., 2022) 4639–4654, [arXiv:2201.05607](#).

O. Friedrich, A. **Halder**, A. Boyle, C. Uhlemann, D. Britt, S. Codis, D. Gruen, and C. Hahn, “The PDF perspective on the tracer-matter connection: Lagrangian bias and non-Poissonian shot noise,” *MNRAS* **510** no. 4, (Mar., 2022) 5069–5087, [arXiv:2107.02300](#).

- A. **Halder**, O. Friedrich, S. Seitz, and T. N. Varga, “The integrated three-point correlation function of cosmic shear,” *MNRAS* **506** no. 2, (Sept., 2021) 2780–2803, arXiv:2102.10177.
- R. Marco Figuera, B. Pham Huu, A. P. Rossi, M. Minin, J. Flahaut, and A. **Halder**, “Online characterization of planetary surfaces: PlanetServer, an open-source analysis and visualization tool,” *Planet. Space Sci.* **150** (Jan., 2018) 141–156.